

COURSE NOTES

Lie Theory

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Chapter 1

Examples of Lie algebras

Lie algebras were originally introduced by Sophus Lie in the 1870s to study the concept of continuous transformation groups. The book *Lie Algebras* by Nathan Jacobson which was published in 1962 is the earliest textbook for Lie algebras, and it was the first time that the terminology “Lie algebra” appears. Before that, people called them infinitesimal Heisenberg groups.

Definition 1.1. *A Lie algebra is simple if it is not abelian and has no nontrivial ideals.*

Where do Lie algebras come from?

1.1 From associative algebras

Denote by $\mathbb{F}$ the ground field. Let A be an associative algebra. $A^{(-)}$ is a Lie algebra by bracket

$$[a, b] = ab - ba \quad \text{for all } a, b \in A.$$

Theorem 1.2 (Wedderburn Theorem). *Suppose $\mathbb{F}$ is algebraically closed.*

- Any simple finite dim. associative algebra is isomorphic to $M_n(\mathbb{F})$.*
- For any finite dim. associative algebra A , there exists a unique nilpotent ideal $N \triangleleft A$ such that $A/N \simeq M_{n_1}(\mathbb{F}) \oplus \dots \oplus M_{n_k}(\mathbb{F})$*

Example 1.3. $\mathfrak{sl}_n$ is simple if $\text{char } \mathbb{F} = 0$ or $\text{char } \mathbb{F} \nmid n$; otherwise $\mathfrak{sl}_n$ is not simple. In the case of $\text{char } \mathbb{F} \mid n$, $\mathbb{F}I_n$ is a 1-dim ideal.

1.2 From involutions

Definition 1.4. Let A be any linear algebra. A linear transformation $*$: $A \rightarrow A$ is called an involution if for any $a, b \in A$,

1. $(a^*)^* = a$;
2. $(ab)^* = b^*a^*$

Example 1.5. 1. Transposition $(a_{ij}) \mapsto (a_{ji})$ is an involution.

2. Symplectic involution for quaternion:

$$\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$$

with the relations

$$i^2 = j^2 = k^2 = -1, \quad ij = k, \quad jk = i, \quad ki = j.$$

The involution is defined by $a + bi + cj + dk \mapsto a - bi - cj - dk$. On the other hand, quaternion can be represented by matrices

$$\begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix},$$

where $a, b, c, d \in \mathbb{R}$ and i is the square root of unity. Then the involution has the form

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \mapsto \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}.$$

Given an involution of an associative algebra A , we have

$$K = \{a \in A : a^* = -a\} \subset A^{(-)}.$$

Then K is a Lie subalgebra.

1.3 From derivations of linear algebras

Definition 1.6. Let A be an associative algebra. A linear map $\alpha : A \rightarrow A$ is called a derivation if for any $a, b \in A$, $\alpha(ab) = \alpha(a)b + a\alpha(b)$.

The commutator of two derivations is again a derivation, i.e., $\text{Der } A \subset \text{End}_{\mathbb{F}}(A)^{(-)}$.

Example 1.7. Let $W_n = \text{Der } \mathbb{F}[x_1, \dots, x_n]$, where

$$\text{Der } \mathbb{F}[x_1, \dots, x_n] = \left\{ \sum_{i=1}^n f_i \frac{\partial}{\partial x_i} : f_i \in \mathbb{F}[x_1, \dots, x_n] \right\}.$$

Note that W_n is an infinite dim Lie algebra (also simple).

Example 1.8. Let A be an associative algebra. Fix $a \in A$, then $\text{ad } a : x \mapsto [a, x]$ is a derivation. More generally, let L be a Lie algebra. Then $\text{ad } L \triangleleft \text{Der } L$ because $[d, \text{ad } x] = \text{ad}(d(x))$ for any $d \in \text{Der } L$ and $x \in L$. The derivations in $\text{ad } L$ are called inner derivations.

Remark 1.9. The appearance of derivations is natural in the sense that for any $d \in \text{Der } A$, if $\exp d := \text{Id} + d + \frac{d^2}{2!} + \dots$ makes sense, then $\exp d$ is an automorphism with inverse $\exp(-d)$. For example, A is finite dimensional or d acts locally nilpotently.

1.4 From brackets

Many algebraic structures contain brackets. Some of them are Lie brackets.

Example 1.10. Let A be an associative commutative algebra. A bracket $[\ , \] : A \times A \rightarrow A$ is called a Poisson bracket if

1. $(A, [\ , \]) is a Lie algebra;$
2. $[a^2, b] = 2a[a, b]$ for any $a, b \in A$.

Remark 1.11. In general, we require $[ac, b] = a[b, c] + [a, b]c$. But in the case of commutative Poisson algebras, this condition is the same as condition 2.

Example 1.12. 1. Let $B = \mathbb{F}(p_1, \dots, p_n; q_1, \dots, q_n)$. For any $f, g \in B$, define

$$[f, g] = \sum_{i=1}^n \left(\frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} - \frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} \right).$$

$B = \mathbb{F}[t]$. For any $f, g \in B$, define $[f, g] = f'g - fg'$.

1.5 From Lie groups

The tangent space at unity of a Lie group is a Lie algebra.

Chapter 2

Engel's Theorem

2.1 A stronger version of Engel's Theorem

Definition 2.1. An associative algebra A is called an enveloping algebra of a Lie algebra L if $L \subset A^{(-)}$ and $A = \langle L \rangle$.

Theorem 2.2. Let A be a finite dim enveloping algebra of L . If $\exists m \geq 1$ such that $\forall a \in L, a^m = 0$, then A is nilpotent.

We will prove a stronger version.

Definition 2.3. Let L be a Lie algebra. We say a subset $S \subset L$ is a Lie subset if $\forall a, b \in S, [a, b] \in S$.

Theorem 2.4. Let A be a finite dim enveloping algebra of L and $S \subset L$ be a Lie subset of L with $L = \text{span } S$. If $\exists m \geq 1$ s.t. $\forall a \in S, a^m = 0$, then A is nilpotent.

Proof. Denote $\dim A = n$. By Zorn's Lemma¹, we can choose a maximal Lie subset $S_1 \subset S$ s.t. the associative algebra $\langle S_1 \rangle$ is nilpotent.² We claim that $S_1 = S$. Otherwise, there exists $s \in S \setminus S_1$ such that $[S_1, s] \subset S_1$.³ Now let $S_2 = S_1 \cup \{s\}$, a Lie subset of S . We will deduce a contradiction by showing the nilpotency of $\langle S_2 \rangle$. Then our claim follows. Hence, we conclude that $A = \langle L \rangle = \langle S \rangle$ is nilpotent.

¹Zorn's lemma is the following statement: Let $(P, \leq)$ be a partially ordered set. If every chain in P has an upper bound in P , then P contains a maximal element.

²Firstly, the set $P = \{S' \subset S : \langle S' \rangle \text{ is nilpotent}\}$ is nonempty by our assumption. For an arbitrary chain

$$Q_1 \subset Q_2 \subset \dots$$

in P , we need to prove the union of all Q_i 's, denoted by Q , is also in P . Let $a_1 a_2 \dots a_l$ be a monomial in $\langle Q \rangle$. There exists $N \in \mathbb{N}$ such that $a_i \in Q_N$ for all i . Note that for any i we have $Q_i^n = 0$. Thus, $a_1 a_2 \dots a_l = 0$ as long as $l \geq n$. Then $Q^n = 0$ as required.

³If $\exists x_1 \in S_1$ s.t. $[x_1, s] \notin S_1$, then take $s_1 = [x_1, s]$. If there still exists $x_2 \in S_1$ s.t. $[x_2, s_1] \notin S_1$, then take $s_2 = [x_2, s_1]$. Iterating, we have $s_i = [x_i, [x_{i-1}, \dots, [x_1, s]]] \in (\text{ad } S_1)^i(s)$. Because $\langle S_1 \rangle$ is nilpotent, $(\text{ad } S_1)^{2n-1} = 0$. Thus, after finite steps, the procedure ends, i.e., $\exists k$ s.t. $[S_1, s_k] \subset S_1$ as desired.

Let us prove $\langle S_2 \rangle$ is nilpotent. Let x be an arbitrary nonzero monomial in $\langle S_2 \rangle$. Then x can be written as

$$x = s_{11}s_{12}\dots s_{1m_1}s^{k_1}s_{21}s_{22}\dots s_{2m_2}s^{k_2}\dots,$$

where $s_{ij} \in S_1$, $m_i < n$ and $k_i < m$. Since $[S_1, s] \subset S_1$, we can move all s to the right of s_{ij} without changing the number of elements in S_1 . For example, $s^{k_1}s_{21} = s^{k_1-1}[s, s_{21}] + s^{k_1-1}s_{21}s$, in which each term in both sides has one element in S_1 . Thus, x have at most n s_{ij} in the expression. Thus, x has length at most $n - 1 + n(m - 1) = mn$. As a consequence, $S_2^{mn} = 0$. $\square$

2.2 Three types of Engel's Theorem

Definition 2.5. Let L be a Lie algebra and M be a vector space over $\mathbb{F}$. Define an operation $L \times M \rightarrow M; (a, m) \mapsto a \cdot m$. The following are equivalent:

1. $L + M$ is a Lie algebra if we define $a \cdot m = -m \cdot a$ and $m_1 \cdot m_2 = 0$ for any $a \in L$ and $m_1, m_2 \in M$.
2. $[a, b] \cdot m = a \cdot (b \cdot m) - b \cdot (a \cdot m)$ for any $a, b \in L$ and $m \in M$.
3. $L \rightarrow (\text{End}_{\mathbb{F}} M)^{(-)}$ is a Lie algebra homomorphism.

We say M is a module of L if it satisfies any of these conditions.

There are three versions of Engel's Theorem:

Theorem 2.6 (Engel's Theorem). Let L be a finite dim Lie algebra. If $\forall a \in L, \text{ad } a$ is nilpotent, then L is nilpotent.

Theorem 2.7. Let L be a Lie algebra and M be a finite dim L -module. If any element in L acts nilpotently, then L acts on M nilpotently.

Theorem 2.8. Let L be a Lie algebra and M be a finite dim L -module. Let $S \subset L$ be a Lie subset with $\text{span } S = L$. If any $a \in S$ acts nilpotently, then L acts nilpotently.

Proof of Theorem 2.6. Let L have dimension n . Then $\forall a \in L, (\text{ad } a)^n = 0$. Consider the subalgebra $\langle \text{ad } L \rangle \subset \text{End}_{\mathbb{F}} M$ generated by $\text{ad } L$. This theorem follows from Theorem 2.2. $\square$

Remark 2.9. We have the following relations:

$$\text{Theorem 2.4} \implies \text{Theorem 2.8} \implies \text{Theorem 2.7} \implies \text{Theorem 2.6}.$$

1. Theorem 2.6 is actually a special case of Theorem 2.7 by taking M as the adjoint representation.

2. Theorem 2.8 is a stronger version of Theorem 2.7.

3. Theorem 2.8 follows directly from Theorem 2.4 by regarding $\varphi(S)$ as a Lie subset of the Lie algebra $\varphi(L)$.

Remark 2.10. Theorem 2.7 and Theorem 2.8 do not require L to be finite dim. But Theorem 2.6 requires because it views L as an L -module.

Corollary 2.11. Let L be a Lie algebra and M be a finite dim L -module is a finite dim representation. If any element in L acts nilpotently, then there exists a basis of M such that L consists of strictly upper triangular matrices; that is,

$$L \subseteq \begin{pmatrix} 0 & * & \cdots & * \\ 0 & 0 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

Proof. Let φ be the representation. By Theorem 2.7, $\varphi(L)$ is nilpotent. Let n be such that $\varphi(L)^n \neq 0$ and $\varphi(L)^{n+1} = 0$. Take a nonzero vector v_1 in $\varphi(L)^n(M)$. Then repeating the procedure for $M/\mathbb{C}v_1$ (not necessarily an L -module), on which $\varphi(L)$ still acts nilpotently gives $v_2 \in M \setminus (\mathbb{C}v_1)$. Iterating, we get a basis $v_1, v_2, \dots, v_{\dim M}$ as desired. $\square$

Chapter 3

Lie's Theorem

Lemma 3.1. *Let $\text{char } \mathbb{F} = 0$ or $\text{char } \mathbb{F} > n$. Let a, b be two $n \times n$ matrices s.t. $[a, [a, b]] = 0$. Then $[a, b]$ is nilpotent.*

Proof. Since $\text{ad } a$ is a derivation, we have

$$(\text{ad } a)^i(b^j) = \begin{cases} 0 & \text{if } j < i, \\ n![a, b]^n & \text{if } j = i. \end{cases}$$

Let $f = x^n + \beta_{n-1}x^{n-1} + \dots + \beta_1x + \beta_0$ be an annihilating polynomial of b . Applying $(\text{ad } a)^n$ to $f(b) = 0$, we have

$$(\text{ad } a)^n(f(b)) = n![a, b]^n = 0.$$

Since $n! \neq 0$, $[a, b]^n = 0$. □

3.1 Two equivalent statement of Lie's Theorem

Theorem 3.2 (Lie's Theorem). *Let L be a solvable Lie algebra over $\mathbb{F}$ and M be a finite dim L -module. Let $\text{char } \mathbb{F} = 0$ or $\text{char } \mathbb{F} > \dim_{\mathbb{F}} M$ and F be algebraically closed. Then there exists a basis of M such that*

$$L \subset \begin{pmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & * \end{pmatrix}.$$

Let us show an equivalent theorem.

Theorem 3.3. *If L is a solvable Lie algebra and M is an irreducible finite dim L -module, then M is 1-dim.*

Proof of Theorem 3.3. As we are mainly concerned with the image of $L \rightarrow (\text{End}_{\mathbb{F}} M)^{(-)}$, we may assume $L \subset (\text{End}_{\mathbb{F}} M)^{(-)}$. Since L is solvable, we have

$$L \supsetneq L^{(1)} \supsetneq L^{(2)} \supsetneq \dots \supsetneq L^{(s)} \supsetneq L^{(s+1)} = 0, \quad \text{for some } s < \dim M.$$

If $s = 0$, i.e., L is abelian, then M contains a common eigenvector v for all elements in L .¹ Since $\mathbb{C}v$ is L -invariant, by irreducibility of M , we get $M = \mathbb{C}v$. Then we are done.

If $s > 0$, then for any $a \in L^{(s)}$ and $b \in L$, $[a, [a, b]] \in [L^{(s)}, L^{(s)}] = L^{(s+1)} = 0$.² By Lemma 3.1, we have $[a, b]^n = 0$. Consider a Lie subset $S = \{[a, b] : a \in L^{(s)}, b \in L\}$, in which any element is nilpotent. By Theorem 2.4, $\text{span } S = [L^{(s)}, L]$ is nilpotent. Note that $[L^{(s)}, L]$ is an ideal of L .³ Then $[L^{(s)}, L]M$ is a submodule of M . By the irreducibility of M and nilpotency of $[L^{(s)}, L]$, this submodule must be 0, that is, $[L^{(s)}, L] = 0$. Then for any $x, y \in L^{(s-1)}$, $[x, [x, y]] = 0$. By Lemma 3.1, $[x, y]$ is nilpotent. Applying Theorem 2.4 again, we have $L^{(s)}$ is nilpotent. Then, by the irreducibility of M , we can conclude that $L^{(s)} = 0$ which is a contradiction. $\square$

Now we show the equivalence between Theorem 3.2 and Theorem 3.3.

Proof. Theorem 3.2 $\Rightarrow$ Theorem 3.3 is obvious. Conversely, let M_1 be a minimal submodule. Then consider M/M_1 which has dimension $\dim M - 1$. Then it follows by induction. $\square$

Example 3.4. Let $\text{char } \mathbb{F} = 2$ then $\mathfrak{gl}_2(\mathbb{F})$ is solvable:

$$L^{(1)} = \begin{pmatrix} \alpha & \gamma \\ \beta & \alpha \end{pmatrix}, \quad L^{(2)} = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}, \quad L^{(3)} = 0.$$

3.2 A stronger version of Lie's Theorem

The following we introduce a version of Lie's Theorem which does not require $\mathbb{F}$ to be algebraically closed.

Theorem 3.5. Let L be a solvable Lie algebra over $\mathbb{F}$ and M be a finite dim L -module. Let $\text{char } \mathbb{F} = 0$ or $\text{char } \mathbb{F} > \dim_{\mathbb{F}} M$, Then $[L, L]$ acts nilpotently.

Proof. Denote by $\overline{\mathbb{F}}$ the closure of $\mathbb{F}$. Consider $\overline{L} = L \otimes_{\mathbb{F}} \overline{\mathbb{F}}$ and $\overline{M} = M \otimes_{\mathbb{F}} \overline{\mathbb{F}}$. Then $\overline{M}$ is naturally an $\overline{L}$ -module. Then by Theorem 3.2, $\overline{L}$ can be represented by upper triangular matrices over $\overline{\mathbb{F}}$. Thus, $[\overline{L}, \overline{L}]$ consisting of strictly upper triangular matrices acts on $\overline{M}$ nilpotently. Since the action of $\overline{L}$ is compatible with the original action. $[L, L]$ acts also nilpotently. $\square$

¹Commutative operators have the same eigenspaces.

² $L^{(i)}$ are ideals of L .

³Since $L^{(s)}$ is an ideal of L , $[[L^{(s)}, L], L] \subset [L^{(s)}, L]$.

With the condition of algebraically closed field, we can deduce Theorem 3.3 by Theorem 3.5: Since $[L, L]$ acts nilpotently and M is irreducible, $[L, L] = 0$. Thus, L is abelian. Then the same argument of the $s = 0$ case in the proof of Theorem 3.3 completes the proof.

Remark 3.6. *In conclusion, we have the following relations:*

$$\text{Theorem 3.5} \xrightarrow{\text{alg. closed}} \text{Theorem 3.2} \iff \text{Theorem 3.3}.$$

Chapter 4

Jordan-Chevalley Decomposition

4.1 Solvable ideals and radicals

Let us start with two easy lemmas.

Lemma 4.1. *Let $I \triangleleft L$, and both I and L/I are solvable. Then L is solvable.*

Proof. Let $I \supset I^1 \supset \dots I^{(m)} = 0$ and $L/I \supset (L/I)^{(1)} \supset \dots \supset (L/I)^{(n)} = 0$. Then

$$L^{(n+m)} = (L^{(n)})^{(m)} \subset I^{(m)} = 0.$$

□

Lemma 4.2. *Let I, J be solvable ideals of L . Then $I + J$ is solvable.*

Proof. $(I + J)/I \cong J/(I \cap J)$. Since $J/(I \cap J)$ and I are both solvable, by Lemma 4.1, $I + J$ is solvable. □

Definition 4.3. *Let $\text{Rad } L$ be the maximal solvable ideal of the Lie algebra L .*

In particular, if L is finite dim, $\text{Rad } L$ exists uniquely because of Lemma 4.2. Note that $L/\text{Rad } L$ has no solvable ideals.

Definition 4.4. *A linear transformation φ is called semisimple if it satisfies*

1. $\exists$ a basis in which the matrix of φ is diagonal.
2. $(\varphi - \alpha)^2 v = 0$ if and only if $(\varphi - \alpha)v = 0$.

These two conditions above are equivalent.

4.2 Review of Jordan decomposition

Let $\mathbb{F}$ be algebraically closed, V be a finite dim $\mathbb{F}$ -vector space and $\varphi : V \rightarrow V$ be a linear transformation.

Denote $V_\alpha = \{\text{root vectors w.r.t. } \alpha\} \cup \{0\} = \ker(\varphi - \alpha)^{\dim V}$.

Theorem 4.5. $V = V_{\alpha_1} \oplus \dots \oplus V_{\alpha_r}$, where $\alpha_1, \dots, \alpha_r$ are distinct eigenvalues of φ .

Proof. There exists a polynomial $f(t) = (t - \alpha_1)^{n_1} \dots (t - \alpha_r)^{n_r}$ such that $f(\varphi) = 0$. Denote by

$$g_i = \frac{f}{(t - \alpha_i)^{n_i}}, \quad \text{for } 1 \leq i \leq r.$$

Since they do not have the common roots, $\gcd(g_i : 1 \leq i \leq r) = 1$. Thus, there exist $h_i \in \mathbb{F}[t]$ such that $\sum_{i=1}^r h_i g_i = 1$. Then for any $v \in V$, $v = \sum_i h_i(\varphi) g_i(\varphi) v$, where $h_i(\varphi) g_i(\varphi) v \in V_{\alpha_i}$.

Next, we show the linear independence of V_{α_i} . Let $v_i \in V_i$ such that $v_1 + v_2 + \dots + v_r = 0$. Suppose there exists $v_i \neq 0$. Then we have $(\varphi - \alpha_i)^{n_i} v_i = 0$ and $g_i v_i = 0$ ¹. Since $(t - \alpha_i)^{n_i}$ and g_i are coprime, $v_i = 0$, which is a contradiction. $\square$

Theorem 4.6. φ can be decomposed to $\varphi = \varphi_s + \varphi_n$, where φ_s is semisimple and φ_n is nilpotent. Moreover, there exists a polynomial $P(t) \in \mathbb{F}[t]$ with zero constant term such that $\varphi_s = P(\varphi)$. In particular, $\varphi_n = (1 - P)(\varphi)$.

Proof. Let $V = V_{\alpha_1} \oplus \dots \oplus V_{\alpha_r}$ as in Theorem 4.5. Set φ_s multiplying V_{α_i} by α_i . Then $\varphi - \varphi_s$ is naturally nilpotent.² By *Chinese Remainder Theorem*, there exists a polynomial $P(t) \in \mathbb{F}[t]$ such that

$$P(t) \equiv \alpha_i \pmod{(t - \alpha_i)^{n_i}} \quad \text{for all } i.$$

Furthermore, we can require $P(t)$ has zero constant term: if $\exists \alpha_i = 0$, then we are done; if not, we just add the equation $P(t) \equiv 0 \pmod{t}$. After adding, CRT still holds.³ $\square$

Theorem 4.7. If we further require $[\varphi_s, \varphi_n] = 0$, the decomposition in Theorem 4.6 is unique. We call it *Jordan decomposition*.

Proof. It is obvious that the decomposition in Theorem 4.6 satisfies the requirement. We use the same notations as in Theorem 4.6. Suppose that we have the second decomposition $\varphi = \varphi'_s + \varphi'_n$ satisfies $[\varphi'_s, \varphi'_n] = 0$. Then φ'_s and φ'_n both commute with φ . Since φ_s and φ_n are polynomials of φ , they also commute with φ_s and φ_n . Thus, $\varphi_s - \varphi'_s$ is still semisimple and $\varphi_n - \varphi'_n$ is still nilpotent. As a consequence, $\varphi_s - \varphi'_s = \varphi_n - \varphi'_n$ is both semisimple and nilpotent, which implies $\varphi_s = \varphi'_s$ and $\varphi_n = \varphi'_n$. $\square$

4.3 Compatibility with adjoint representation

We continue to use notations as in Section 4.2. In particular, $\varphi : V \rightarrow V$ is a linear transformation in $\text{End}_{\mathbb{F}} V = \mathfrak{gl}(V)$.

¹ g_i kills v_j for all $j \neq i$.

²Check on each V_{α_i} .

³ t has no common roots with $(t - \alpha_i)^{n_i}$'s.

Theorem 4.8. *If $\varphi = \varphi_s + \varphi_n$ is a Jordan decomposition, then $\text{ad } \varphi = \text{ad } \varphi_s + \text{ad } \varphi_n$ is a Jordan decomposition. In particular, $\text{ad } \varphi_s$ and $\text{ad } \varphi_n$ can be expressed by polynomials of $\text{ad } \varphi$ with zero constant terms.*

Proof. 1. $\text{ad } \varphi_s$ is semisimple, because the action on the basis element E_{ij} satisfies

$$\left[\begin{pmatrix} \alpha_1 & & \\ & \alpha_2 & \\ & & \ddots \\ & & & \alpha_r \end{pmatrix}, E_{ij} \right] = (\alpha_i - \alpha_j)E_{ij}.$$

2. $\text{ad } \varphi_n$ is nilpotent because $(\text{ad } \varphi_n)^{\dim V} = 0$.

3. $[\text{ad } \varphi_s, \text{ad } \varphi_n] = \text{ad}[\varphi_s, \varphi_n] = 0$ since ad is a Lie algebra homomorphism. □

Chapter 5

Cartan's Criteria

In this chapter, we always assume that $\mathbb{F}$ has characteristic 0 and V is a finite dim vector space over $\mathbb{F}$.

5.1 Replicas

Definition 5.1. Let φ be a linear transformation on V . Suppose there is a basis of V such that

$$\varphi = \begin{pmatrix} \alpha_1 & & & \\ & \alpha_2 & & \\ & & \ddots & \\ & & & \alpha_n \end{pmatrix}.$$

We call a linear transformation ψ a replica of φ if

$$\psi = \begin{pmatrix} \beta_1 & & & \\ & \beta_2 & & \\ & & \ddots & \\ & & & \beta_n \end{pmatrix} \quad \text{and} \quad \alpha_i = \alpha_j \Rightarrow \beta_i = \beta_j.$$

Then we have two easy properties of replicas.

Proposition 5.2. If ψ is a replica of φ , then $\exists f \in \mathbb{F}[t]$ s.t. $\psi = f(\varphi)$.

Proof. By Chinese Remainder Theorem, we have a solution f for the system of equations:

$$f(t) \equiv \beta_i \pmod{t - \alpha_i}, \quad \text{for all } i.$$

Then f is as desired. □

Proposition 5.3. Let φ be a semisimple linear transformation defined as in Definition 5.1 and $\pi \in \text{Hom}_{\mathbb{Q}}(\mathbb{F}, \mathbb{Q})$ be a $\mathbb{Q}$ -linear function. Let $\psi = \pi(\varphi) = \text{diag}(\pi(\alpha_1), \dots, \pi(\alpha_n))$, then $\text{ad } \psi$ is a replica of $\text{ad } \varphi$.

Proof. In the proof of Theorem 4.8, we have proved that $\text{ad } \psi$ and $\text{ad } \varphi$ are semisimple with eigenvalues $\alpha_i - \alpha_j$'s and $\pi(\alpha_i) - \pi(\alpha_j)$'s, respectively. Moreover, if $\alpha_i - \alpha_j = \alpha_p - \alpha_q$, then $\pi(\alpha_i) - \pi(\alpha_j) = \pi(\alpha_p) - \pi(\alpha_q)$. $\square$

Then we have an immediate corollary by Proposition 5.2.

Corollary 5.4. *Let φ and ψ be as in Proposition 5.3. Then $\exists f \in \mathbb{F}[t]$ s.t. $\text{ad } \psi = f(\text{ad } \varphi)$.*

5.2 Cartan's Criterion for solvability

Recall that V is finite dimensional.

Lemma 5.5 (Nilpotency criterion). *Let $A \subseteq B \subseteq \mathfrak{gl}(V)$ be subspaces. Denote by $M = \{\varphi \in \mathfrak{gl}(V) : [\varphi, B] \subseteq A\}$. If $\varphi \in M$ and $\text{tr}(\varphi M) = 0$, then φ is nilpotent.*

Remark 5.6. *Why M is better than L ? Let φ and ψ be as in Proposition 5.3. Then*

$$\varphi \in M \iff \text{ad } \varphi : B \rightarrow A \xrightarrow{\text{Cor 5.4}} \text{ad } \psi : B \rightarrow A \iff \psi \in M.$$

Proof of Lemma 5.5. Let $\varphi = \varphi_s + \varphi_n$ be the Jordan decomposition of φ . Take an arbitrary linear transformation $\pi \in \text{Hom}_{\mathbb{Q}}(\mathbb{F}, \mathbb{Q})$ and let $\psi = \pi(\varphi_s)$. Under some basis, φ_s and ψ can be expressed as a matrix

$$\varphi_s = \begin{pmatrix} \alpha_1 & & \\ & \ddots & \\ & & \alpha_n \end{pmatrix} \quad \text{and} \quad \psi = \begin{pmatrix} \pi(\alpha_1) & & \\ & \ddots & \\ & & \pi(\alpha_n) \end{pmatrix}.$$

By Remark 5.6, $\psi \in M$, then $\text{tr}(\varphi\psi) = 0$. This gives $\text{tr}(\varphi_s\psi) = 0$, because ψ commutes with φ_n (by Proposition 5.2) and $\varphi_n\psi$ is nilpotent. Thus, we have $\sum_i \alpha_i \pi(\alpha_i) = 0$. Applying π again, we get a sum of rational numbers $\sum_i \pi(\alpha_i)^2 = 0$, which means $\pi(\alpha_i) = 0$ for all i . By arbitrariness of π , we finally get $\varphi_s = 0$, that is, $\varphi = \varphi_n$ is nilpotent. $\square$

Theorem 5.7 (Cartan's Criterion for solvability). *If a Lie algebra $L \subseteq \mathfrak{gl}(V)$ satisfies $\text{tr}(L[L, L]) = \{0\}$, then L is solvable.*

Proof. Let $M = \{x \in \mathfrak{gl}(V) : [x, L] \subseteq [L, L]\}$ (take $A = [L, L]$, $B = L$ in Lemma 5.5).

We will prove $\text{tr}([a, b]M) = 0$ for any $a, b \in L$. Let $x \in M$. Note that

$$\text{tr}([a, b]x) = \text{tr}([ax, b]) - \text{tr}(a[x, b]) = 0,$$

because the elements in $[L, L]$ has trace 0 and $[x, b] \in [L, L]$.

By Lemma 5.5 we can deduce that $[a, b]$ acts nilpotently for any $a, b \in L$. Namely, the elements in the Lie subset $S = \{[a, b] : a, b \in L\}$ act nilpotently. Then by Theorem 2.4, we have $[L, L] = \text{span } S$ acts nilpotently, which implies L is solvable. $\square$

5.3 Cartan's Criterion for semisimplicity

Recall that for a finite dimensional Lie algebra L , $\text{Rad } L$ is the largest solvable ideal and contains any solvable ideal of L . Moreover, the quotient Lie algebra $L/\text{Rad } L$ does not contain any nonzero solvable ideals. We say a Lie algebra is semisimple if its radical is 0.

Definition 5.8 (Killing form). *We call the bilinear form defined as $(a|b) = \text{tr}(\text{ad } a \text{ ad } b)$ for all $a, b \in L$ the Killing form of L .*

Killing form has associativity: Let $x, y, z \in \mathfrak{gl}(V)$. $\text{tr}([x, y]z - x[y, z]) = \text{tr}([x, y]z + \text{ad } x[z, y]) = \text{tr}([xz, y]) = 0$. In particular, $\text{ad } L \subseteq \mathfrak{gl}(L)$.

Theorem 5.9 (Cartan's Criterion for semisimplicity). *For a finite dimensional Lie algebra L , L is semisimple if and only if its Killing form is nondegenerate.*

Proof. Suppose L is semisimple. Let $I = \{a \in L : (a|L) = 0\}$. Since the Killing form is associative, $I \triangleleft L$. Note that $\text{tr}(\text{ad } I \text{ ad } I) \subseteq (I|I) = \{0\}$. By Theorem ??, $\text{ad } I \subseteq \mathfrak{gl}(L)$ is solvable. Consider the Lie algebra homomorphism

$$\text{ad}|_I : I \rightarrow \text{ad } I; a \mapsto \text{ad } a.$$

We have $\text{ad } I = I/\ker = I/(I \cap Z(L))$, where $Z(L)$ is the center of L . Since $\text{ad } I$ and $I \cap Z(I)$ are solvable, I is solvable.

Suppose the Killing form is nondegenerate and $J \triangleleft L$ is a nonzero solvable ideal of L . Then L contains a nonzero abelian ideal I .¹ For any $a \in I$ and $b \in L$, we have $(\text{ad } a \text{ ad } b)^2 = 0$, since $[b, [a, [b, L]]] \in I$. Thus, $(a|b) = \text{tr}(\text{ad } a \text{ ad } b) = 0$, which is a contradiction. $\square$

¹Let $J^{(n)} \neq 0$ and $J^{(n+1)} = 0$. Then $J^{(n)}$ is the desired ideal.

Chapter 6

Consequences of Cartan's Criteria

In this chapter we always assume L is finite dimensional.

6.1 Decomposition of semisimple Lie algebras

Let us first review Linear Algebra.

Proposition 6.1. *Let V be a vector space with a symmetric bilinear form $(\mid)$ and $W \subseteq V$ be a finite dimensional subspace on which $(\mid)_W$ is nondegenerate. Denote by $W^\perp = \{v \in V : (v|W) = \{0\}\}$. Then $V = W \oplus W^\perp$.*

Proof. It is obvious that $W \cap W^\perp = \{0\}$. Let $w_1, w_2, \dots, w_n$ be a basis of W . For any $v \in V$, we have a system of equations

$$(v - \alpha_1 w_1 - \alpha_2 w_2 - \dots - \alpha_n w_n | w_j) = 0, \quad \text{for } j = 1, \dots, n,$$

where α_i 's are viewed as variables. Since $\det(w_i | w_j) \neq 0$, the system has a unique solution, also denoted by α_i . Thus, $v - \sum_i \alpha_i w_i \in W^\perp$ as desired. $\square$

Remark 6.2. *If we require V is finite dimensional and $(\mid)_V$ is nondegenerate, we can only deduce $\dim V = \dim W + \dim W^\perp$.*

Proposition 6.3. *If L is semisimple and $(0) \neq I \triangleleft L$, then $(\mid)_I$ is nondegenerate.*

Proof. Assume that $\exists$ nonzero $a \in I$ s.t. $\text{tr}(\text{ad } a \text{ ad } I) = \{0\}$. Then

$$([a, I] | L) = (a | [I, L]) \subseteq (a | I) = \{0\},$$

which implies $[a, I] = (0)$. Then $I \cap C_L(I) \neq (0)$. For any $b \in I \cap C_L(I)$, we have $([b, L] | I) = (b | [L, I]) = (b | I) = \{0\}$. Thus, $I \cap C_L(I)$ is an abelian ideal of L . This is a contradiction with semisimplicity of L . $\square$

Theorem 6.4. *If L is semisimple, then L is a direct sum of simple ideals. Moreover, this decomposition is unique.*

Proof. Let I be an ideal of L . Then by Proposition 6.3 and Proposition 6.1, $L = I \oplus I^\perp$. Note that $I^\perp$ is also an ideal of L by associativity of the Killing form. Then by induction on $\dim L$, we obtain the desired decomposition

$$L = I_1 \oplus I_2 \oplus \dots \oplus I_r,$$

where I_i are simple ideals of L .¹

To show the uniqueness, let I be a simple ideal of L . Then $[I, L] = I$ because $Z(L) \subseteq \text{Rad}(L) = (0)$. On the other hand, $[I, L] = \bigoplus_i [I, I_i]$. Since I is simple, $[I, I_i] \triangleleft I$ is I itself or (0) . So all but one summand must be (0) . Say $[I, I_i] = I$. Then $I = I_i$. $\square$

The following is a corollary of Proposition 6.3 and Cartan's Criterion of semisimplicity.

Corollary 6.5. *Every ideal of a semisimple Lie algebra L is still semisimple. Moreover, every quotient of L is still semisimple.*

Proof. The first statement follows immediately from Proposition 6.3 and Theorem 5.9. Let $I \triangleleft L$ be an ideal. Then by Proposition 6.1, $L/I \cong I^\perp$. By the associativity of the Killing form, $I^\perp$ is also an ideal of L , and thus semisimple. $\square$

6.2 Derivations

Recall that inner derivations are derivations in $\text{ad } L$.

Theorem 6.6. *L is semisimple, then all derivations are inner.*

Proof. Let $d : L \rightarrow L$ be a derivation. Consider the Lie algebra $\tilde{L} := L \oplus \mathbb{F}d$, whose brackets are defined as

$$[d, a] = d(a) = -[a, d].$$

It is straight to check it is a well-defined Lie algebra. Note that $\tilde{L}$ is not semisimple. Otherwise, by Theorem 6.4, $\tilde{L} = L \oplus (1\text{-dim ideal})$, which is a contradiction.

Then $\text{Rad } \tilde{L} \neq 0$. Say $a + d \in \text{Rad } \tilde{L}$ for some $a \in L$. For any $b \in L$, since $\text{Rad } \tilde{L}$ is an ideal, $[a + d, b] = [a, b] + d(b) \in \text{Rad } \tilde{L} \cap L$. Since $\text{Rad } \tilde{L} \cap L$ is a solvable ideal in L , it must be $\{0\}$. Thus, $d = -\text{ad } a$. $\square$

Example 6.7. *There are examples of outer derivations (derivations but not inner).*

1. *Let A be a commutative associative algebra. Then $A^{(-)}$ is an abelian Lie algebra and all inner derivations are 0. Thus, all nonzero derivations are outer.*

¹Actually we need to show if $J \triangleleft I \triangleleft L$ be an ideal, J is an ideal of L . This holds because $I^\perp$ is an ideal.

2. Let $A = \text{Mat}(2; \mathbb{F}[X])$. Consider the Lie algebra $A^{(-)}$ and a derivation defined by

$$\begin{pmatrix} f(X) & g(X) \\ h(X) & u(X) \end{pmatrix}' = \begin{pmatrix} f'(X) & g'(X) \\ h'(X) & u'(X) \end{pmatrix}.$$

Note that it is NOT an inner derivation: Otherwise, this inner derivative maps all constant matrix(whose elements are constants) to 0. This means a matrix commutes with all constant matrix. It can only be zero matrix. Moreover, it is obvious that this derivation is not a zero map. Hence, we deduce a contradiction.

Chapter 7

Casimir Elements of L -Modules

7.1 Tensor products of modules

Proposition 7.1. *If V and W are two L -modules, there is a well-defined L -module structure on $V \otimes_{\mathbb{F}} W$ by*

$$a(v \otimes w) = av \otimes w + v \otimes aw, \quad \text{for all } v \in V, w \in W.$$

Proof. Define $\varphi : V \times W \rightarrow V \otimes W$ by $\varphi(v, w) = av \otimes w + v \otimes aw$, which is a well-defined bilinear map. Thus, by universal property of tensor products, $\exists!$ $\tilde{\varphi}$ s.t. the following diagram commutes.

$$\begin{array}{ccc} V \times W & \xrightarrow{\varphi} & V \otimes W \\ & \searrow & \uparrow \tilde{\varphi} \\ & & V \otimes W \end{array}$$

Thus, the action is well-defined. □

Definition 7.2. *If V is an L -module, then $V^* = \text{Hom}_{\mathbb{F}}(V, \mathbb{F})$ has an L -module structure defined by $[a, \varphi]v = -\varphi(av)$.*

Furthermore, we can define L -module structure on $\text{Hom}_{\mathbb{F}}(V, W)$ by $[a, \varphi]v = a\varphi(v) - \varphi(av)$, where $\varphi \in \text{Hom}_{\mathbb{F}}(V, W)$. It is a special case of above proposition if we regard $\text{Hom}_{\mathbb{F}}(V, W)$ as $V^* \otimes W$.

7.2 Schur's Lemma

Lemma 7.3. *Let $\mathbb{F}$ be an algebraically closed field and V be a finite dim vector space. Let $\Sigma \subset \text{End}_{\mathbb{F}}(V)$, and we assume that Σ acts on V irreducibly (no nontrivial invariant space). Then the centralizer $C(\Sigma) = \{\varphi \in \text{End}_{\mathbb{F}}(V) : \varphi a = a\varphi, \forall a \in \Sigma\}$ consists of scalar multiplications.*

Proof. Note that $C(\Sigma)$ is an associative algebra of $\text{End}_{\mathbb{F}}(V)$. We will prove it actually is a division algebra. Let $0 \neq c \in C(\Sigma)$. Note that both $\ker c$ and $\text{im } c$ are Σ -invariant. Then $\ker c = (0)$ and $\text{im } c = V$, which implies c is invertible. Moreover, c^{-1} satisfies $c^{-1}\varphi = c^{-1}\varphi cc^{-1} = \varphi c^{-1}$. Thus, $c^{-1} \in C(\Sigma)$.

Lemma 7.4. *The only finite dim division algebra over algebraically closed field $\mathbb{F}$ is $\mathbb{F}$ itself.*

proof of Lemma 7.4. Let C be a division algebra and $c \in C$. Denote $\dim_{\mathbb{F}} C = d$. Then the $d+1$ elements $1, c, \dots, c^d$ are linearly dependent. $\exists f \in \mathbb{F}[t]$, say $f(t) = \alpha(t - \alpha_1) \dots (t - \alpha_r)$, s.t. $f(c) = \alpha(c - \alpha_1 1) \dots (c - \alpha_r 1) = 0$. Since C is a division algebra, which is naturally an integral domain, $\exists i$ s.t. $c = \alpha_i 1$. □

Then our claim follows immediately from Lemma 7.4. □

The following is a consequence of Schur's Lemma.

Corollary 7.5. *If L is a finite dim simple Lie algebra, any associative nondegenerate symmetric bilinear form must be a scalar multiple of the Killing form.*

Proof. Let $(\cdot | \cdot)_1$ and $(\cdot | \cdot)_2$ be two associative nondegenerate symmetric bilinear forms of L . Since L is finite dimensional and the forms are nondegenerate, $\exists \alpha \in \text{End}_{\mathbb{F}}(L)$ s.t.

$$(u | \cdot)_1 = (\alpha(u) | \cdot)_2, \quad \text{for any } u \in L.$$

By associativity, we have for any $u, v \in L$

$$(\alpha([u, v]) | \cdot)_2 = ([u, v] | \cdot)_1 = (u | [v, \cdot])_1 = (\alpha(u) | [v, \cdot])_2 = ([\alpha(u), v], \cdot)_2.$$

It implies that α commutes with $\text{ad } v$ for any $v \in L$. By Lemma 7.3, $\alpha \in C(\text{ad } L)$. Since L is simple, namely $\text{ad } L$ acts irreducibly, α must be a scalar multiplication. □

7.3 Casimir operators

In this section, we always assume L is a finite dimensional semisimple Lie algebra. Let $(\cdot | \cdot)$ be an associative nondegenerate symmetric form. Note that $(\cdot | \cdot)$ is not necessarily the Killing form. But by Corollary 7.5, in each simple component, it is a scalar multiple of the Killing form.

Let $e_1, e_2, \dots, e_n$ be a basis of L and $e^1, e^2, \dots, e^n$ be the dual basis with respect to $(\cdot | \cdot)$, i.e. $(e_i | e^j) = \delta_{ij}$. Denote $c = \sum_i e_i \otimes e^i \in L \otimes_{\mathbb{F}} L$ which is called the Casimir element of L .

Proposition 7.6. *View $L \otimes L$ as a tensor product of L -modules. $L \cdot c = 0$.*

Proof. We need the following lemma.

Lemma 7.7. For any $a, b \in L$, $(\sum_i e_i \otimes e^i | a \otimes b) = (a | b)$.

Proof. Let $a = \sum_i \alpha_i e^i$ and $b = \sum_i \beta_i e_i$. Then

$$(\sum_i e_i \otimes e^i | a \otimes b) = \sum_{i=1}^n (e_i | a) (e^i | b) = \sum_{i=1}^n \alpha_i \beta_i = (a | b).$$

□

For any $a \in L$, we need to show that $a \cdot c = \sum_{i=1}^n [a, e_i] \otimes e^i + \sum_{i=1}^n e_i \otimes [a, e^i] = 0$. Since $(\cdot | \cdot)_{L \otimes L}$ is nondegenerate¹, it suffices to show that for any $b \otimes d \in L \otimes L$, $(a \cdot c | b \otimes d) = 0$.

$$\begin{aligned} \left(\sum_{i=1}^n [a, e_i] \otimes e^i + \sum_{i=1}^n e_i \otimes [a, e^i] \middle| b \otimes d \right) &= \sum_{i=1}^n ([a, e_i] | b) (e^i | d) + \sum_{i=1}^n (e_i | b) ([a, e^i] | d) \\ &= - \sum_{i=1}^n (e_i | [a, b]) (e^i | d) - \sum_{i=1}^n (e_i | b) (e^i | [a, d]) \\ &= -([a, b] | d) - (b | [a, d]) \\ &= 0. \end{aligned}$$

□

Remark 7.8. Note that c does not depend on the choice of $e_1, \dots, e_n$. If there is another basis $f_1, \dots, f_n$, then the bilinear form of $\sum_i e_i \otimes e^i - \sum_i f_i \otimes f^i$ with every $a \otimes b \in L \otimes L$ is zero. Since the bilinear form is nondegenerate, these two summations coincide.

Let A be an associative algebra satisfying $L \subseteq A^{(-)}$ and L generates A , i.e., A is an associative enveloping algebra of L . We can regard $LL \subseteq A$ as an L -module by $a \cdot bc := [a, bc] = [a, b]c + b[a, c] \in LL$. Note that there exists an L -module homomorphism from $L \otimes L$ to LL by $b \otimes c \mapsto bc$. We call the image of c under this homomorphism (also denoted by c) the Casimir operator of A . Proposition 7.6 tells us that $c = \sum_i e_i e^i \in Z(A)$.

¹We can check by its Gram matrix. Let $A = (a_{ij})$ be the Gram matrix of $(\cdot | \cdot)_V$ under the basis $x_1, \dots, x_n$. Then under the basis $x_1 \otimes x_1, \dots, x_1 \otimes x_n, x_2 \otimes x_1, \dots, x_n \otimes x_n$, the Gram matrix of $(\cdot | \cdot)_{L \otimes L}$ is

$$\begin{pmatrix} a_{11}A & a_{12}A & \cdots & a_{1n}A \\ a_{21}A & a_{22}A & \cdots & a_{2n}A \\ \vdots & \vdots & & \vdots \\ a_{n1}A & a_{n2}A & \cdots & a_{nn}A \end{pmatrix},$$

which has determinant $|A|^{2n} \neq 0$.

Chapter 8

Complete Reducibility

8.1 Completely reducible modules

Let $\mathbb{F}$ be a field (the characteristic and algebraical closeness are irrelevant). Let A be a linear algebra (not necessarily containing unity) and V be an A -module (not necessarily finite dimensional).

Proposition 8.1. *The following two conditions are equivalent:*

1. $\forall$ submodule $W \subseteq V$ is a direct summand, $\exists$ a submodule $W' \subseteq V$ s.t. $V = W \oplus W'$.
2. $V = \bigoplus_{i \in I} V_i$, where V_i 's are irreducible.

Proof. $1 \Rightarrow 2$. We claim that it is enough to show that $V = \sum_{i \in I} V_i$, for some index set I . Suppose we have such summation. Let I' be a maximal subset of I such that $\sum_{i \in I'} V_i$ is direct. The existence of I' is due to Zorn's Lemma. Then for any V_j , $j \in I \setminus I'$, $(\bigoplus_{i \in I'} V_i) \cap V_j = (0)$ or V_j by irreducibility of V_j . By the maximality of I' , the intersection must be V_j . Thus, $V = \bigoplus_{i \in I'} V_i$.

Now I need to show that $W := \sum(\text{irreducible submodules}) = V$. If not, then $\exists$ nonzero submodule $W' \subseteq V$ s.t. $V = W \oplus W'$. Say $w \in W'$ and $w \neq 0$. Denote by P the submodule generated by w . Note that we have a maximal submodule $Q \subseteq P$.¹ Suppose $V = Q \oplus Q'$. One can prove that $P' := Q' \cap P$ satisfies $P = P' \oplus Q$. Then $P' \cong P/Q$. By the maximality of Q , P' is irreducible, which is a contradiction with our assumption to W .

$2 \Rightarrow 1$. For any $W \subsetneq V = \bigoplus_i V_i$, we denote by I' the maximal subset of I s.t. $(\bigoplus_{i \in I'} V_i) \cap W = (0)$.² We claim $V = (\bigoplus_{i \in I'} V_i) + W$. It suffices to show that it contains any V_j . If $\exists j \in I$ s.t. V_j is not contained, then

$$\left(\bigoplus_{i \in I' \cup \{j\}} V_i \right) \cap W = (0).$$

¹Since any proper module does not contain w , the union of them does not still contain w . Then by Zorn's Lemma, a maximal submodule $Q \subseteq P$ exists.

²The existence is due to Zorn's Lemma.

This is a contradiction with the maximality of I' . $\square$

Definition 8.2. *An A -module satisfying either of the equivalent conditions in Proposition 8.1 is called completely reducible.*

8.2 Weyl's theorem

Compared with the generality of the last section, in this section, we assume L is a finite dimensional semisimple Lie algebra over an algebraically closed field $\mathbb{F}$ of characteristic 0.

Theorem 8.3. *Every finite dimensional L -module is completely reducible.*

Proof. Let V be a finite dimensional L -module. It suffices to show for any submodule $W \neq V$ a submodule of V as its direct sum complement. The proof is divided into two parts.

Part I: W is irreducible and has codim 1. Note that the bilinear form $(a|b) := \text{tr}(a|_W b|_W)$ is nondegenerate symmetric and associative. Symmetry and associativity can be proved similar to the Killing form. Suppose $I \triangleleft L$ is an ideal s.t. $(I|L) = \{0\}$. Then $(I|[I]) = \{0\}$. By Cartan's Criterion (Theorem 5.7), the image of I is solvable. By Corollary 6.5, the image of I is semisimple, which is a contradiction.

Let $e_i, 1 \leq i \leq n$ be a basis of L and e^i be its dual basis w.r.t. $(\cdot | \cdot)_W$. Then $c = \sum_i e_i e^i$ satisfies $\text{tr}(c|_W) = \sum_i (e_i | e_i) = n$. On the other hand, by Proposition 7.6, $c|_W$ lies in the centralizer of $L|_W$. By Schur's Lemma, $c|_W = \alpha \cdot \text{Id}_W$ for some constant $\alpha \in \mathbb{F}$. Then $\alpha = n / \dim W \neq 0$.

The quotient V/W is a 1-dim L -module. Thus, $L = [L, L]$ acts as 0 on V/W .³ This implies that $L \cdot V \subseteq W$. In particular, $c \cdot V = W$. So we have $V = W \oplus \ker c$. Part I is completed.

Part II: W is reducible and has codim 1. We use induction by the dimension of V , thanks to Part I. Suppose $W' \subset W$ is a proper submodule. Then $W/W' \subseteq V/W'$ also has codim 1. Then by induction, $\exists v \in V$ s.t.

$$V/W' = (\mathbb{F}v + W')/W' \oplus W/W'$$

with $\dim(\mathbb{F}v + W')/W' = 1$. Obviously, $v \notin W'$ and $v \notin W$. By induction hypothesis again, $\mathbb{F}v + W' = W' \oplus (1\text{-dim } L\text{-module})$. Moreover, this 1-dim L -module is not contained in W since v does not belong to W . By a routine check, we have $V = (1\text{-dim } L\text{-module}) \oplus W$.

Part III: W is general. $\square$

³ $\forall x \in L, x$ acts as a scalar on V/W . Thus, $[L, L]$ acts as 0.